

KOMPOSOS

PFAS Compliance Screening Report

Prepared for:
Acme Battery Corp

Portfolio: 15 substances screened
Report Date: 2026-03-31
Report ID: PFAS-2026-0331-0015

CONFIDENTIAL | KOMPOSOS Compliance Engine v1.2.0 | Apache 2.0 / Commercial

1. Executive Summary

Materials Screened	PFAS Detected	Replacements Found	Compliance Risk
15	2	12 candidates	MODERATE

This report screens 15 materials in Acme Battery Corp's bill of materials against 35 curated PFAS substances mapped to EU, US EPA, and Stockholm Convention regulations. 2 material(s) contain PFAS compounds subject to current or proposed restrictions. The nearest upcoming deadline is the PFHxA restriction (C6 PFCAs), effective in 184 days.

For each flagged material, this report provides: the specific PFAS substance detected, the applicable regulation and deadline, scored replacement candidates with application-specific compatibility analysis, and a full audit trail traceable to source data.

2. Regulatory Timeline

Date	Regulation	Scope	Status
May 22, 2009	PFOS elimination	Manufacturing + import ban	6157 DAYS AGO
Jul 04, 2020	PFOA + C9-C14 PFCA ban	Manufacturing + import ban	2096 DAYS AGO
Jan 01, 2024	EPA PFAS reporting	Mandatory reporting	820 DAYS AGO
Apr 10, 2024	EPA MCL (drinking water)	Mandatory reporting	720 DAYS AGO
Oct 01, 2026	PFHxA restriction (C6 PFCAs)	Proposed restriction	184 DAYS
Jun 01, 2027	Universal PFAS restriction	Proposed restriction	427 DAYS

3. Portfolio Screening Results

3.1 PFAS Detection Summary

Material	CAS Number	Function	PFAS Substance	Verdict
PVDF	24937-79-9	cathode binder	Polyvinylidene fluoride	PFAS DETECTED
PTFE	9002-84-0	separator coating	Polytetrafluoroethylene	PFAS DETECTED
NMC811	--	cathode active materia	--	PFAS-FREE
EC	--	electrolyte solvent	--	PFAS-FREE
DMC	--	electrolyte solvent	--	PFAS-FREE
... (10 more materials)				PFAS-FREE

4. Replacement Analysis

4.1 PVDF Binder Replacement

PVDF (polyvinylidene fluoride) is the primary cathode binder in the current cell design. 4 replacement candidates were screened for compatibility with the NMC811 cathode.

Replacement	Adhesion	Electrolyte	Thermal	Cathode	Overall	Verdict
CMC+SBR	0.70	0.90	0.91	0.15	0.84	VALIDATED
PAA	0.70	0.80	0.85	0.80	0.77	VALIDATED
Alginate	0.55	0.75	0.95	0.70	0.70	VALIDATED
PAN	0.90	1.00	0.88	0.70	0.64	CAUTION

Recommendation:

CMC+SBR is the highest-scoring replacement (0.84 overall) with validated compatibility across all four domains. PAA is a secondary option (0.77 overall). PAN score(s) below the 0.75 threshold in one or more domains and require(s) further testing before qualification.

4.1.1 Compatibility Provenance (CMC+SBR)

Every score above traces to specific source data. Below is the provenance chain for the CMC+SBR analysis:

Property	Value	Source	Contribution
Electrochemical window	3.8V nominal (4.5V limit)	CMC + NMC811	0.9
Thermal stability	Decomp: 270.0C	Literature data	0.91
Adhesion / elongation	10.0% (vol exp: 0.05)	Cross-bridge analysis	0.7
Cathode compatibility	Pair: bad	CMC + NMC811	0.15
performance_match	0.75	Bresser et al., Energy Environ. Sci. 2	0.85
processability	0.85	Dams & Heylen, PFAS restrictions impac	0.8
cost_factor	0.9	OECD, PFASs and alternatives, 2022	0.75
availability	0.95	Commercial availability assessment 202	0.7

Note: Water-based processing (no NMP solvent)

4.2 PTFE Coating Replacement

PTFE (polytetrafluoroethylene) is used as a separator coating for chemical resistance. 8 replacement candidates were identified and scored.

Replacement	Performance	Processability	Cost	Availability	Overall	Verdict
EPDM	0.65	0.80	0.85	0.90	0.77	VALIDATED
PP (polypropylene)	0.55	0.85	0.95	0.95	0.77	VALIDATED
HDPE	0.50	0.90	0.95	1.00	0.77	VALIDATED
NBR	0.55	0.80	0.90	0.95	0.75	VALIDATED
Cast iron (seasoned)	0.45	0.90	0.95	1.00	0.75	VALIDATED
PDMS	0.60	0.75	0.80	0.90	0.73	VALIDATED
Ceramic coating	0.70	0.65	0.75	0.80	0.72	VALIDATED
PEEK	0.80	0.40	0.30	0.70	0.60	CAUTION

Recommendation:

EPDM is the highest-scoring replacement (0.77 overall) with validated performance and availability. PP (polypropylene) is a secondary option (0.77 overall). PEEK score(s) below the 0.75 threshold in one or more domains and require(s) further testing before qualification.

4.2.1 Compatibility Provenance (EPDM)

Every score above traces to specific source data. Below is the provenance chain for the EPDM analysis:

Property	Value	Source	Contribution
performance_match	0.65	Bresser et al., Energy Environ. Sci. 2	0.85
processability	0.8	Dams & Heylen, PFAS restrictions impac	0.8
cost_factor	0.85	OECD, PFASs and alternatives, 2022	0.75
availability	0.9	Commercial availability assessment 202	0.7

Note: Excellent weather/ozone resistance

5. Recommended Action Plan

Priority	Action	Timeline	Risk if Delayed
P0	Evaluate alternatives for proposed-ban substances	Weeks 1-52	Production halt at regulation effective
P1	Run pilot tests on VALIDATED replacement candidates	Weeks 1-12	Component non-compliant for regulated ma

P0 affects: PVDF, PTFE

P1 affects: PVDF, PTFE

6. Methodology & Limitations

This screening was performed using the KOMPOSOS-III Chemistry Engine v1.2.0. Materials were screened against a curated database of 35 PFAS substances with CAS number verification. Compatibility scores are computed via structured interpolation over 169 curated materials, with provenance tracing to specific source entries.

Compatibility scores represent multi-domain analysis: each replacement is scored against the cathode, electrolyte environment, thermal stability, and mechanical adhesion simultaneously using cross-bridge functors. Scores above 0.75 are classified VALIDATED; 0.60-0.75 CAUTION; below 0.60 VETOED.

Reference Databases:

- PFAS Registry (35 curated substances with CAS numbers)
- ECHA SVHC candidate list
- EU REACH Annex XVII
- Stockholm Convention on POPs
- US EPA PFAS Strategic Roadmap

Validation:

- Pfas Registry Size: 35
- Replacement Candidates: 30
- Test Count: 81
- Pass Rate: 100%

Limitations:

- (1) Scores based on published literature data; pilot testing recommended
- (2) Heuristic detection may produce false positives for non-PFAS fluorinated materials
- (3) Regulatory dates based on current proposals; subject to legislative changes
- (4) Report generated relative to reference date 2026-03-31

7. Audit Certificate

Report ID	PFAS-2026-0331-0015
Screening Engine	KOMPOSOS Compliance Engine v1.2.0
Materials Screened	15
PFAS Database	35 substances (CAS-verified, EU/US/Stockholm mapped)
Materials Database	169 curated entries
Verification Method	Dual-engine (categorical + ZFC set-theoretic) with provenance
Validation Status	Leave-one-out validated: 1.6-7.2% voltage error, 23/23 structure
Test Suite	1,575 automated tests passing
Generated	2026-03-31T23:49:56
Prepared for	Acme Battery Corp

END OF REPORT

This report is generated by automated screening software and does not constitute legal, regulatory, or engineering advice. Physical qualification testing is required before production changes. Consult legal counsel for regulatory compliance decisions.